

Topology-Preserving Image Segmentation with Spatial-Aware Persistent Feature Matching

By

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DECLARATION

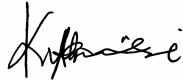
This work was carried out at AIMS Rwanda in partial fulfilment of the requirements for a Master of Science Degree.

I hereby declare that except where due acknowledgement is made, this work has never been presented wholly or in part for the award of a degree at AIMS Rwanda or any other University.

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Abstract

In biomedical imaging, road extraction, and crack detection, where several target structures are thin, interconnected, and sensitive to small errors, image segmentation is crucial. Even a little break in an artery or worm structure can result in an incorrect structural evaluation in biological images. This thesis investigates spatial-aware persistent feature matching for topology-preserving image segmentation. In order to preserve connected components, branching, and holes, the work combines topology-aware information from persistent homology with pixel-wise learning. By formulating predictions that stay essentially close to the ground truth, the goal is to improve segmentation quality beyond pixel accuracy.

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List of Abbreviations

- **BCE** – Binary Cross-Entropy
- **BM** – Betti Matching
- **CNN** – Convolutional Neural Network
- **clDice** – Centerline Dice
- **DRIVE** – Digital Retinal Images for Vessel Extraction
- **FOV** – Field of View
- **GT** – Ground Truth
- **PH** – Persistent Homology
- **PD** – Persistence Diagram
- **ReLU** – Rectified Linear Unit
- **RGB** – Red, Green, Blue
- **SATLoss** – Spatial-Aware Topological Loss
- **TDA** – Topological Data Analysis
- **U-Net** – U-shaped Convolutional Neural Network

1. Introduction

Image segmentation, which allocates pixels to relevant categories, is an essential step in computer vision. It is commonly utilized in applications where both pixel-level accuracy and structural correctness are crucial, such as biomedical imaging, road extraction, and crack identification. A right segmentation result in these applications must preserve the object structure in addition to separating the object from the background. Through encoder–decoder architectures and skip connections, deep learning techniques like U-Net have enhanced image segmentation. However, a lot of segmentation techniques still use pixel-based loss functions such binary cross-entropy and dice loss. Although these losses increase local pixel overlap, structural errors, which can be significant when working with thin or connected structures, are not explicitly taken into account [1, 2, 3].

The issue that a segmentation result may be extremely accurate at the pixel level yet still be topologically erroneous is the main emphasis of this thesis. For instance, a road may be divided at an intersection, a crack may split into multiple parts, or an expected blood vessel may have a slight break. Even though these mistakes might only impact a small number of pixels, they have the power to alter an object's true structure by producing erroneous holes, missing connections, or false components [4, 3].

To address this problem, this study investigates spatial-aware persistent feature matching for topology-preserving image segmentation. The idea is to include structural information in the learning process instead of relying only on pixel correspondence. Persistent homology is useful in this setting because it can capture connected components and holes across different thresholds, allowing the topology of the predicted mask to be compared with the topology of the ground truth [5, 4, 6].

Earlier topology-aware methods showed the importance of persistent homology, but many of them mainly compare topological features globally. This can be limited because two masks may have the same number of connected components or holes while placing them in different image locations. Therefore, spatial information is important for improving the matching between predicted and ground-truth topological features. Recent works on induced matching and spatial-aware persistent feature matching address this issue by combining topological similarity with spatial consistency [7, 8]. This thesis therefore considers a segmentation framework that combines pixel-level learning, persistent feature extraction, and spatial-aware topological matching. The intended contribution is to support segmentation that is accurate at the pixel level while also preserving important structural and topological features.

1.1 Related Work

Image segmentation has been thoroughly researched since the introduction of deep learning algorithms. By directly learning image features from data, convolutional neural networks have enhanced segmentation. U-Net is a significant model that retains spatial information by using skip connections and an encoder–decoder architecture [9]. Because of this, U-Net performed well

in pixel-level segmentation tasks and medical imaging. To enhance segmentation performance in various applications, other U-Net variations have also been introduced. However, the majority of these techniques still depend mainly on pixel-level losses, which increase precision but don't necessarily preserve object structure.

The failure of pixel-based techniques to preserve topology is a significant drawback. Images of vessels, cracks, roadways, or other thin structures can have significant structural changes due to small local errors. Because traditional loss functions do not take into account global structural aspects, Mosinska et al. demonstrated that they are inadequate for topology-aware segmentation [2]. The necessity to go beyond pixel-wise losses was revealed by this effort. Similarly, by emphasizing connectedness rather than just region overlap, cLDice was developed to enhance tubular structure segmentation [3]. Although it does not adequately explain all topological phenomena, such as holes, its primary benefit is that it helps in the preservation of centerline structures.

Topological data analysis has also been used to address structural errors in segmentation. Persistent homology provides a way to represent connected components and holes across different scales [5, 10]. Its advantage is that it gives a mathematical description of object structure, which can be used to compare the topology of prediction and ground truth. Hu et al. were among the first to integrate persistent homology with deep learning for topology-preserving image segmentation [4]. Clough et al. also introduced a topological loss that uses prior knowledge about the expected object structure to guide the learning process [6]. These methods improve structural preservation, but many of them mainly compare topology globally.

Recent work has focused on improving how topological structures are matched. Stucki et al. proposed Betti matching, which improves the alignment between true and predicted structures using barcode matching [7]. This improves topological comparison, but spatial location is still important because two features may have similar birth and death values while appearing in different image regions. Wen et al. addressed this by introducing spatial-aware persistent feature matching, which includes the location of topological features in the matching process [8]. This makes the matching more meaningful for segmentation tasks. These developments show that combining topology with spatial information can provide a stronger framework for structure-preserving image segmentation.

1.2 Motivation

Pixel precision is insufficient for topology-sensitive image segmentation. A tiny break in an artery or worm structure in biomedical images can have an impact on how the whole structure is evaluated. Similar to this, missing connections can divide a single continuous structure into multiple disconnected components in road extraction and crack detection. Consequently, preserving structure is crucial, particularly for thin and interconnected structures [2, 3].

Although deep learning models frequently perform well at the pixel level, their predictions may nevertheless have disconnected components, misleading holes, or broken branches. This causes an error between structural correctness and pixel-wise precision. Such structural features can be

measured with the use of topological data analysis. Specifically, related components and holes across various thresholds can be represented by persistent homology [5].

However, topology alone may not be sufficient because two features can have similar birth and death values but appear in different image locations. This motivates the use of spatial-aware persistent feature matching, which combines topological similarity with spatial consistency. This thesis is therefore motivated by the need to produce segmentation results that are not only accurate in pixels, but also structurally meaningful.

1.3 Objectives

The main objective of this thesis is to study topology-preserving image segmentation using spatial-aware persistent feature matching, with the aim of improving both pixel-level accuracy and structural consistency.

To achieve this, the study is guided by the following objectives:

- To examine the limitations of pixel-wise loss functions in preserving thin and connected structures.
- To study how persistent homology can capture important topological features such as connected components and holes.
- To develop a segmentation framework that combines deep learning with topology-aware losses, including cIDice and SATLoss.
- To evaluate the framework on topology-sensitive datasets and compare its performance with BCE-only training.

2. Topology-Aware Image Analysis

This chapter introduces the main ideas needed to understand topology-aware image segmentation. It begins with basic image segmentation concepts, then explains how topology describes connected components and holes in images. The chapter also introduces persistent homology, cubical complexes, persistence diagrams, Wasserstein distance, and spatial-aware persistent feature matching. These concepts provide the mathematical foundation for comparing the structure of a predicted mask with the ground truth.

2.1 Image Segmentation Fundamentals

2.1.1 Definition of Image Segmentation. Image segmentation is the process of assigning a label to each pixel in an image. The aim is to separate the object of interest from the background while keeping its real shape and structure. This task is important in medical imaging, biological analysis, road extraction, and other computer vision applications [1, 9].

2.1.2 Binary Segmentation and Pixel-wise Prediction. In binary segmentation, each pixel is classified as either background or foreground. Let I be the input image and M be the ground-truth mask. Then

$$M(x, y) \in \{0, 1\},$$

where 0 represents background and 1 represents the target object. Deep learning models usually treat this as a pixel-wise classification problem, where each pixel is predicted from features learned from the image [9].

2.1.3 Probability Maps and Thresholding. Most segmentation models first produce a probability map $P(x, y) \in [0, 1]$, where $P(x, y)$ gives the probability that pixel (x, y) belongs to the foreground. A binary mask is then obtained by applying a threshold τ :

$$\widehat{M}(x, y) = \begin{cases} 1, & \text{if } P(x, y) \geq \tau, \\ 0, & \text{otherwise.} \end{cases}$$

Here, $\widehat{M}$ denotes the predicted binary mask. Figure 2.1 illustrates this process from input image to final binary mask.

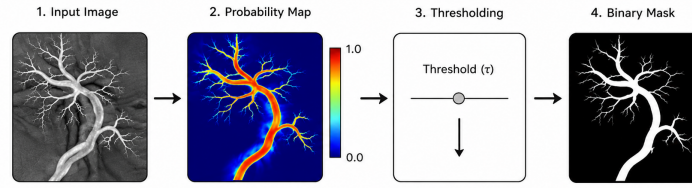


Figure 2.1: Basic image segmentation pipeline. The model takes an input image, produces a probability map, applies a threshold τ , and gives the final binary mask.

A small change in $P(x, y)$ near the threshold can change the final mask, so thresholding can affect the final structure of the segmentation.

2.1.4 Structural Challenges in Segmentation. Although deep learning models can give high pixel accuracy, they may still fail to preserve object structure. This is common in thin, elongated, or connected components. For example, a small hole can split one vessel, road, or crack into two disconnected components, even when most pixels are correctly classified.

This shows the limitation of relying only on pixel-wise accuracy. Traditional losses mainly compare pixels or regions, but they do not directly check whether the prediction preserves connected components, holes, or branches [2, 3]. This motivates the use of topological methods, which are introduced in the following sections.

2.2 Topology in Image Analysis

2.2.1 Basic Topological Concepts. Topology studies features of a shape that remain important even when the shape is stretched or slightly deformed. In image segmentation, this is useful because a prediction may have high pixel accuracy but still break a connection or create a false hole. Such errors change the structure of the object, especially in thin structures such as roads, vessels, and cracks [10, 5].

Connected component. A connected component is a region whose pixels form one continuous object. For example, one unbroken road or vessel should usually appear as one connected component.

Hole. A hole is an enclosed empty region inside an object. In segmentation, a hole may be a real structure or a false hole created by prediction error.

2.2.2 Betti Numbers. Betti numbers count important topological features. For two-dimensional images, the most useful Betti numbers are

$$\beta_0 = \text{number of connected components}, \quad \beta_1 = \text{number of holes}.$$

Here, β_0 counts separate objects, while β_1 counts enclosed holes [5].

Example 2.1: Betti Numbers in Binary Masks.

A mask with two separate objects and no hole has

$$\beta_0 = 2, \quad \beta_1 = 0.$$

A mask with one object and one enclosed hole has

$$\beta_0 = 1, \quad \beta_1 = 1.$$

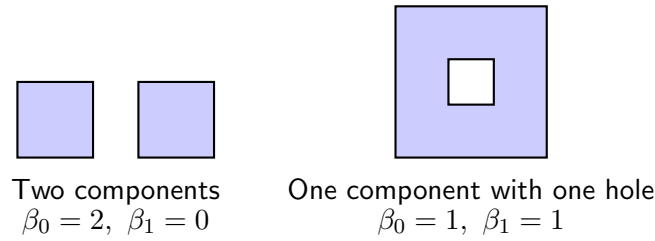


Figure 2.2: Examples of Betti numbers in binary images.

These examples show why topology is important in segmentation. A good model should not only predict correct pixels, but also preserve the main connected components and holes. This idea leads naturally to persistent homology, which studies how these topological features appear and disappear across different thresholds.

2.3 Persistent Homology

Persistent homology is a tool from topological data analysis used to study how the structure of a space changes across different thresholds. In image segmentation, it tracks when connected components and holes appear and disappear in a probability map [5, 10]. This makes it useful for measuring whether a predicted mask preserves the structure of the ground truth [4, 6].

2.3.1 Filtration Process. Let $P(x, y) \in [0, 1]$ be the probability map produced by a segmentation model. The threshold is applied directly to the probability map. For a given threshold t , we keep only the pixels whose probability is at least t . This gives the set

$$X_t = \{(x, y) \in \Omega : P(x, y) \geq t\},$$

where Ω is the image domain.

This means that X_t is the foreground region obtained at threshold t . For example, if $t = 0.8$, only highly confident foreground pixels are included. If $t = 0.4$, more pixels are included. Therefore, as the threshold decreases, the foreground region grows:

$$X_{t_1} \subseteq X_{t_2} \subseteq \cdots \subseteq X_{t_k}, \quad t_1 > t_2 > \cdots > t_k.$$

This nested sequence is called a filtration. It allows us to observe how the topology of the predicted object changes from high-confidence pixels to lower-confidence pixels [5, 4].

Example 2.2: Thresholding a Probability Map.

Consider a simple probability map

$$P = \begin{bmatrix} 0.9 & 0.8 & 0.1 \\ 0.2 & 0.7 & 0.6 \\ 0.1 & 0.3 & 0.5 \end{bmatrix}.$$

At threshold $t = 0.8$, the foreground set is

$$X_{0.8} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

At threshold $t = 0.6$, more pixels are included:

$$X_{0.6} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

At threshold $t = 0.5$, another pixel is added:

$$X_{0.5} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Thus,

$$X_{0.8} \subseteq X_{0.6} \subseteq X_{0.5}.$$

Persistent homology keeps track of the appearance and disappearance of connected components and holes through analyzing each thresholded mask.

2.3.2 Birth and Death of Features. A topological feature is *born* when it first appears during filtration and dies when it merges, disappears, or fills in [5]. Birth implies the appearance of a new, distinct region for a connected component, whereas death implies the joining of an older component. When a hole is born, an enclosed empty area appears; when it dies, the hole is filled.

Each feature is written as

$$(b, d),$$

where b is its birth value and d is its death value. Its persistence is

$$\text{pers}(b, d) = |b - d|.$$

A large persistence means the feature is more stable, while a small persistence usually indicates noise [10].

Example 2.3. If a component is born at $b = 0.6$ and merges with an older component at $d = 0.5$, then its persistence pair is

$$(0.6, 0.5),$$

with

$$\text{pers} = |0.6 - 0.5| = 0.1.$$

This means the component lived only briefly during the filtration.

2.4 Cubical Complexes from Images

A digital image is a grid of pixel values. To study its topology, this grid is represented as a cubical complex, where connected components and holes can be computed. A cubical complex is suitable for images because it is built from grid elements such as vertices, edges, and squares [5, 11].

A grayscale image can be represented as a function

$$I : \Omega \subset \mathbb{Z}^2 \rightarrow \mathbb{R},$$

where Ω is the set of pixel locations and $I(x, y)$ is the intensity value at pixel (x, y) .

Example 2.4: Grayscale Image as a Matrix.

A grayscale image can also be written as a matrix of pixel values. For example,

$$I = \begin{bmatrix} 5 & 1 & 6 \\ 2 & 9 & 4 \\ 8 & 3 & 7 \end{bmatrix}.$$

Each entry represents the intensity value of one pixel in the image grid.

2.4.1 Cells and Image Construction. In two dimensions, a cubical complex is made of

$$0\text{-cells} = \text{vertices}, \quad 1\text{-cells} = \text{edges}, \quad 2\text{-cells} = \text{squares}.$$

Vertices are grid points, edges connect neighbouring vertices, and squares represent filled pixel regions. Together, these cells allow the image to be treated as a topological space.

There are two common ways to build a cubical complex from an image. In the **V-construction**, image values are assigned to vertices. In the **T-construction**, image values are assigned to square cells, so pixels are treated as filled squares [11].

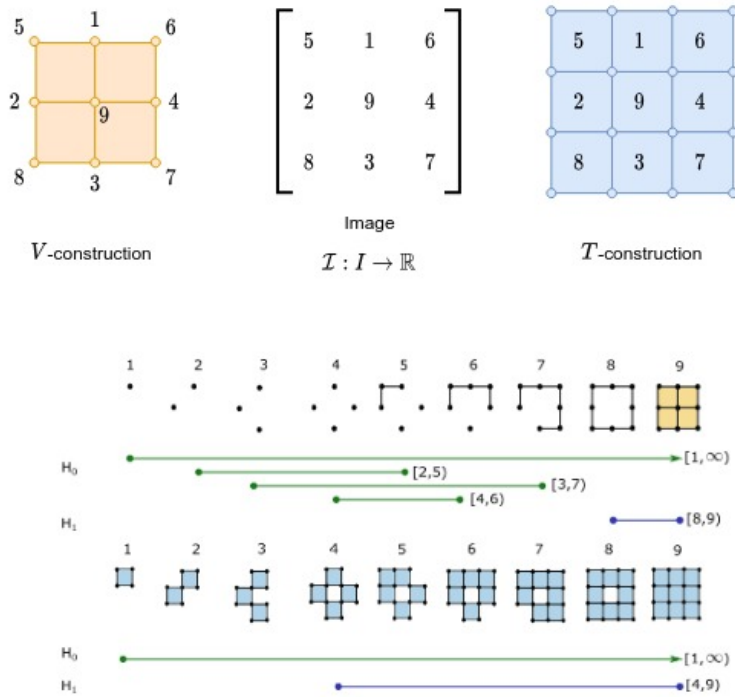


Figure 2.3: V-construction and T-construction for a 3×3 image. In the V-construction, values are placed on vertices, while in the T-construction, values are placed on square pixel cells. The filtration then records how connected components and holes appear or disappear, and the results are summarized by the H_0 and H_1 barcodes. **Adapted from [11].**

2.4.2 Topological Feature Extraction. Figure 2.3 shows how the same image can be converted into cubical filtrations. In the V-construction, vertices appear according to their assigned values, and edges or squares are added when their boundary cells are present. In the T-construction, pixels appear directly as filled squares. Since the two constructions assign image values differently, the birth and death values of features may also differ [11].

The extracted features are summarized using barcodes. The H_0 barcode records connected components, while the H_1 barcode records holes. For example, an interval

$$[8, 9)$$

means that a hole is born at filtration value 8 and dies at value 9. Its persistence is

$$9 - 8 = 1.$$

An interval such as

$$[1, \infty)$$

represents a connected component that is born at value 1 and never dies.

This is important for segmentation because probability maps and masks can be converted into cubical complexes. Their connected components and holes can then be compared between prediction and ground truth during topology-aware learning.

2.5 Persistence Diagrams

A persistence diagram summarizes the topological features obtained from persistent homology. In image segmentation, we usually compare two diagrams: one from the predicted segmentation and one from the ground-truth mask. Let P denote the prediction and M denote the ground truth. Then

$$D(P) = \{(b_i^P, d_i^P)\}_{i=1}^n$$

is the persistence diagram of the prediction, while

$$D(M) = \{(b_j^M, d_j^M)\}_{j=1}^m$$

is the persistence diagram of the ground truth.

Each point (b, d) represents one topological feature, where b is the birth value and d is the death value [5, 10]. Points far from the diagonal have large persistence and usually represent important features. Points close to the diagonal have small persistence and are often less stable or noisy.

More explicitly, the prediction diagram can be written as

$$D(P) = \begin{bmatrix} b_1^P & d_1^P \\ b_2^P & d_2^P \\ \vdots & \vdots \\ b_n^P & d_n^P \end{bmatrix} \in \mathbb{R}^{n \times 2},$$

and the ground-truth diagram can be written as

$$D(M) = \begin{bmatrix} b_1^M & d_1^M \\ b_2^M & d_2^M \\ \vdots & \vdots \\ b_m^M & d_m^M \end{bmatrix} \in \mathbb{R}^{m \times 2}.$$

The first column contains birth values, and the second column contains death values. The numbers of points, n and m , may be different because the prediction and ground truth may contain different numbers of topological features.

Persistence diagrams are important because they allow the topology of the predicted mask and the ground-truth mask to be compared. However, they mainly describe features using birth and death values. They do not directly show where each feature is located in the image. This limitation motivates the spatial-aware matching method discussed later in this chapter [4, 6, 8].

2.5.1 Feature Matching Between Persistence Diagrams. Let $D(M)$ be the persistence diagram of the ground-truth mask and $D(P)$ be the persistence diagram of the predicted mask:

$$D(M) = \{r_i\}_{i=1}^m, \quad D(P) = \{q_j\}_{j=1}^n,$$

where

$$r_i = (b_i^M, d_i^M), \quad q_j = (b_j^P, d_j^P).$$

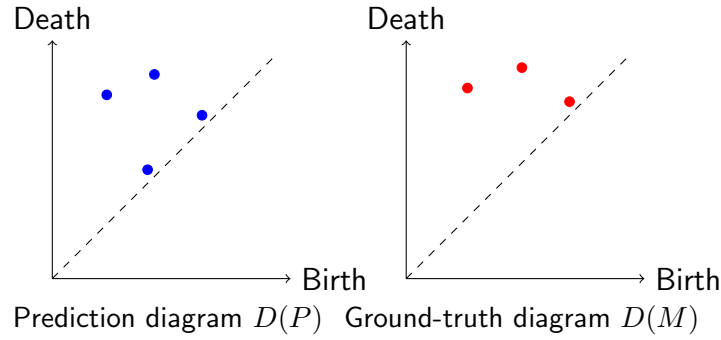


Figure 2.4: Persistence diagrams for prediction and ground truth. Each point represents one topological feature. Comparing $D(P)$ and $D(M)$ helps measure whether the prediction preserves the topology of the ground truth.

Each point represents one topological feature by its birth and death values [5, 10].

To compare the two diagrams, every ground-truth feature is compared with every predicted feature using

$$C_{ij} = \|r_i - q_j\|_p.$$

This produces the pairwise cost matrix

$$C = \begin{array}{c|cccc} & q_1 & q_2 & \cdots & q_n \\ \hline r_1 & C_{11} & C_{12} & \cdots & C_{1n} \\ r_2 & C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ r_m & C_{m1} & C_{m2} & \cdots & C_{mn} \end{array} \in \mathbb{R}^{m \times n}.$$

Each entry C_{ij} measures the cost of matching feature r_i with feature q_j . Small values indicate similar topological features, while large values indicate dissimilar features.

2.6 Wasserstein Distance

The Wasserstein distance compares two persistence diagrams by finding the matching with the smallest total cost. It measures how different the topology of the prediction is from the topology of the ground truth [5, 6].

A matching function η pairs each feature in $D(M)$ with a feature in $D(P)$ or with the diagonal if no suitable match exists. The p -Wasserstein distance is defined as

$$W_p(D(M), D(P)) = \left(\inf_{\eta} \sum_{r_i \in D(M)} \|r_i - \eta(r_i)\|^p \right)^{\frac{1}{p}}.$$

Here,

$$r_i = (b_i^M, d_i^M)$$

is a ground-truth feature, and $\eta(r_i)$ is its matched feature in the prediction.

In simple terms, Wasserstein distance:

computes pairwise costs $\rightarrow$ finds the best matching $\rightarrow$ computes the total matching error.

A small Wasserstein distance means that the prediction preserves the topology of the ground truth well. A large value indicates missing, misplaced, or incorrect topological structures.

Example 2.5: Wasserstein Matching Cost.

Suppose

$$r = (0.90, 0.10) \in D(M), \quad q = (0.80, 0.20) \in D(P).$$

Using Euclidean distance,

$$\|r - q\| = \sqrt{(0.90 - 0.80)^2 + (0.10 - 0.20)^2} \approx 0.141.$$

This small value shows that the predicted feature is close to the ground-truth feature in the persistence diagram.

2.6.1 Matching to the Diagonal. In persistence diagram matching, the first aim is to match a ground-truth feature in $D(M)$ with a similar predicted feature in $D(P)$. For example, if a ground-truth point r_i and a predicted point q_j represent the same connected component or hole, they are matched directly using their birth and death values.

However, the two diagrams may not contain the same number of points. The prediction may create extra features, or it may miss important ground-truth features. In this case, any unmatched feature is matched to the diagonal [5, 6].

The diagonal is defined as

$$\Delta = \{(b, d) \in \mathbb{R}^2 : b = d\}.$$

A point on the diagonal has zero persistence because

$$|b - d| = 0.$$

Therefore, matching a feature to the diagonal means treating it as a feature that should disappear.

For a feature

$$q = (b, d),$$

its projection onto the diagonal is

$$\pi_{\Delta}(q) = \left(\frac{b + d}{2}, \frac{b + d}{2} \right).$$

Features along the diagonal are typically weak or noisy and have little persistence. Because they persist throughout various filtration layers, features that are far away from the diagonal have a great deal of persistence and are hence more significant [10].

Example 2.6: Matching to the Diagonal.

Suppose a predicted feature

$$q = (0.55, 0.50)$$

has no matching feature in the ground truth. Its diagonal projection is

$$\pi_{\Delta}(q) = \left(\frac{0.55 + 0.50}{2}, \frac{0.55 + 0.50}{2} \right) = (0.525, 0.525).$$

The cost of matching q to the diagonal is

$$\|q - \pi_{\Delta}(q)\| = \sqrt{(0.55 - 0.525)^2 + (0.50 - 0.525)^2} \approx 0.035.$$

This small cost shows that q is close to the diagonal, so it is a weak feature. If an important feature far from the diagonal is missing, the matching cost becomes much larger.

2.7 Spatial-Aware Persistent Feature Matching

Ordinary Wasserstein matching compares persistence diagrams using only birth and death values. However, two features may have similar persistence values while appearing in completely different image locations. This can produce incorrect matching in segmentation tasks where spatial position is important [7, 8].

Spatial-aware persistent feature matching solves this problem by combining:

$$\text{topological similarity} + \text{spatial similarity}.$$

2.7.1 Creator and Destroyer Pixels. During filtration, each topological feature appears and later disappears:

- the **creator pixel** is the pixel where the feature is born,
- the **destroyer pixel** is the pixel where the feature dies.

Thus, a feature is represented by

$$(b, d, x, y),$$

where (b, d) are the birth–death values and (x, y) gives the spatial location.

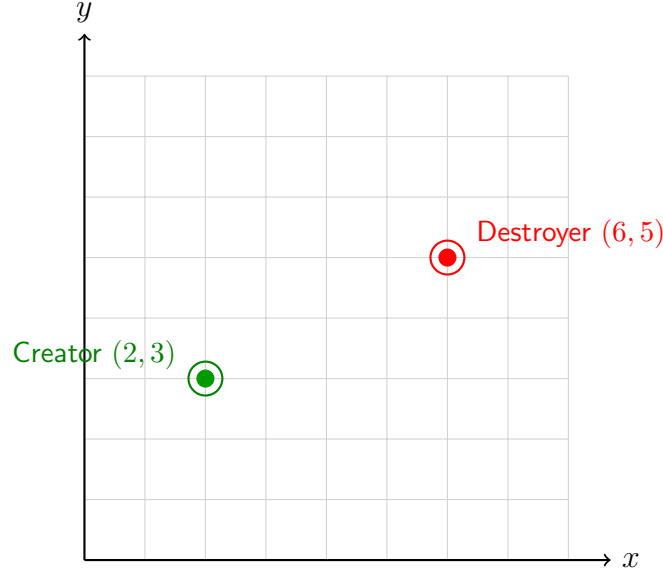


Figure 2.5: Illustration of creator and destroyer coordinates used in spatial-aware persistent feature matching. The creator feature is located at $(2, 3)$ and the destroyer feature at $(6, 5)$.

2.7.2 Persistence and Spatial Distances. Let

$$D(M) = \{r_i\}_{i=1}^m, \quad D(P) = \{q_j\}_{j=1}^n$$

be the persistence diagrams of the ground truth and prediction.

Each feature is written as

$$r_i = (b_i^M, d_i^M), \quad q_j = (b_j^P, d_j^P).$$

The persistence-diagram distance is

$$C_{ij} = \|r_i - q_j\|_p.$$

For the creator coordinates,

$$c_i^M = (x_i^M, y_i^M), \quad c_j^P = (x_j^P, y_j^P).$$

After normalization by the image size p ,

$$\tilde{c}_i^M = \frac{1}{p} c_i^M, \quad \tilde{c}_j^P = \frac{1}{p} c_j^P.$$

The spatial distance becomes

$$S_{ij} = \|\tilde{c}_i^M - \tilde{c}_j^P\|_p.$$

Therefore:

$$\begin{aligned} \text{topological distance} &= \|r_i - q_j\|_p, \\ \text{spatial distance} &= \|\tilde{c}_i^M - \tilde{c}_j^P\|_p. \end{aligned}$$

2.7.3 Spatially Weighted Matching. The persistence distance matrix and spatial distance matrix are combined entrywise:

$$\begin{array}{ccc}
 \text{PD_dist} & \odot & \text{Spatial_dist} \\
 \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} & \odot & \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix} \\
 & = & \\
 & \text{Weighted_dist} & \\
 & \begin{bmatrix} S_{11}C_{11} & S_{12}C_{12} & S_{13}C_{13} \\ S_{21}C_{21} & S_{22}C_{22} & S_{23}C_{23} \\ S_{31}C_{31} & S_{32}C_{32} & S_{33}C_{33} \end{bmatrix} &
 \end{array}$$

Thus, two features obtain a small matching cost only when:

- their persistence values are similar,
- and their spatial locations are also close.

2.7.4 Spatial-Aware Wasserstein Distance. The ordinary Wasserstein distance is

$$W_p(D(M), D(P)) = \left(\inf_{\eta} \sum_{r_i \in D(M)} \|r_i - \eta(r_i)\|_p^p \right)^{\frac{1}{p}}.$$

Spatial-aware matching extends this distance by including both persistence and spatial distances:

$$W_p^{SA}(D(M), D(P)) = \left[\inf_{\eta: D(M) \rightarrow D(P)} \sum_{q \in D(M)} \|q - \eta(q)\|_p^p \|c(q) - c(\eta(q))\|_p^p \right]^{\frac{1}{p}},$$

where

$$\|q - \eta(q)\|_p$$

measures the topological difference between two persistence points, and

$$\|c(q) - c(\eta(q))\|_p$$

measures the spatial distance between their creator coordinates.

This matching strategy preserves both topology and spatial consistency, making it more suitable for topology-sensitive image segmentation tasks [8].

3. Topology-Preserving Segmentation Framework

This chapter outlines the suggested framework for segmentation. The CNN and U-Net components are introduced, and the training losses for pixel accuracy, region overlap, connection, and topology preservation are then explained. Additionally, it summarizes how gradients are passed via creator and destroyer cells during training due to topological loss.

3.1 Deep Learning Components of the Framework

3.1.1 Convolutional Neural Networks. Convolutional Neural Networks (CNNs) are deep learning models designed for image data. They learn useful patterns directly from pixels while preserving spatial relationships in the image [12, 13, 9]. In segmentation, CNNs are used to extract features such as edges, textures, boundaries, and object shapes.

A CNN is mainly built from convolution, activation, and pooling operations. Convolution extracts local patterns, activation introduces nonlinearity, and pooling reduces the spatial size of feature maps while keeping important information.

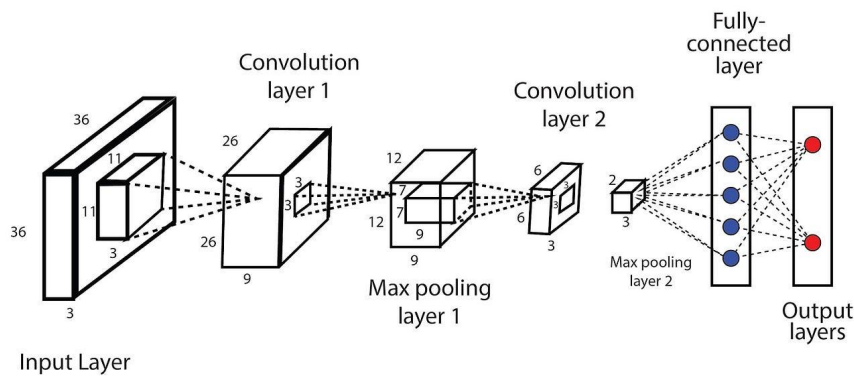


Figure 3.1: Basic CNN pipeline. Convolution extracts features, pooling reduces spatial size, and the output layers use the learned features for prediction.

Convolution Operation

A convolution uses a small matrix called a **kernel** or **filter**. The kernel slides over the image and

computes weighted sums of nearby pixel values. This allows the network to detect local patterns such as edges and boundaries.

Given an input image I and a kernel K , the output feature map F is written as

$$F(x, y) = \sum_i \sum_j K(i, j) I(x - i, y - j).$$

In most CNN implementations, cross-correlation is used, although it is commonly called convolution in deep learning [13].

Example 3.1: Convolution Operation.

For a 3×3 image patch and a 3×3 kernel,

$$I = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 2 & 1 & 0 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix},$$

the output value is

$$(1)(1) + (0)(2) + (-1)(1) + (1)(0) + (0)(1) + (-1)(2) + (1)(2) + (0)(1) + (-1)(0) = 0.$$

This value shows the response of the kernel to that local image region.

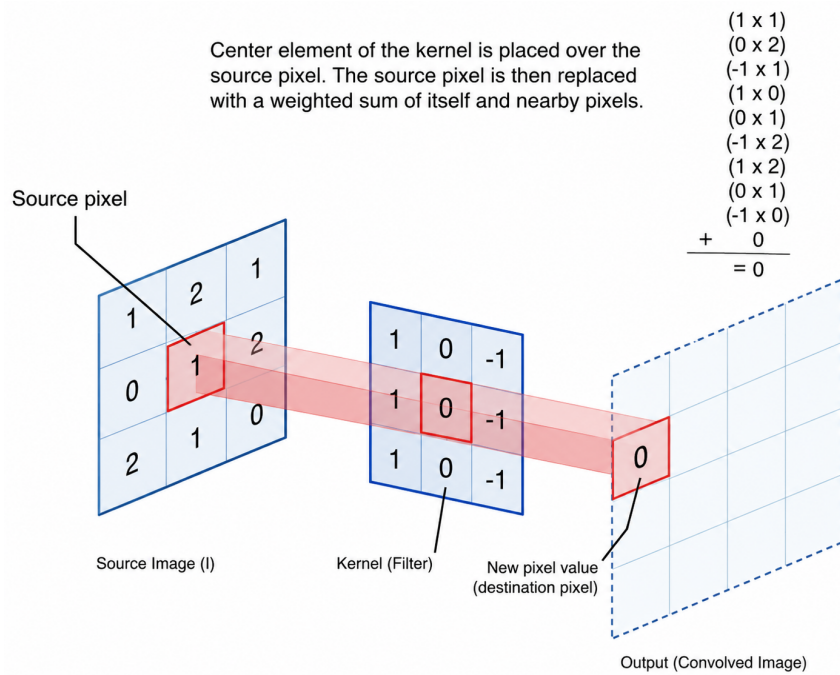


Figure 3.2: Example of a convolution operation. The kernel is applied to an image patch to produce one output value.

After convolution, a nonlinear activation function is usually applied. A common choice is ReLU:

$$\text{ReLU}(z) = \max(0, z).$$

ReLU keeps positive values and sets negative values to zero. This helps the network learn nonlinear image patterns.

Feature Extraction and Pooling

As an image passes through several convolution layers, the network learns features at different levels. Early layers detect simple structures such as edges, while deeper layers combine them into larger shapes and object parts.

Pooling reduces the spatial size of feature maps. This lowers computation and keeps the strongest responses in local regions. Max pooling, for example, selects the largest value from each local window [13].

Example 3.2: Max Pooling Operation.

A 2×2 max pooling operation with stride 2 divides the feature map into non-overlapping 2×2 regions and keeps the maximum value from each region. This reduces a 4×4 feature map to a 2×2 feature map.

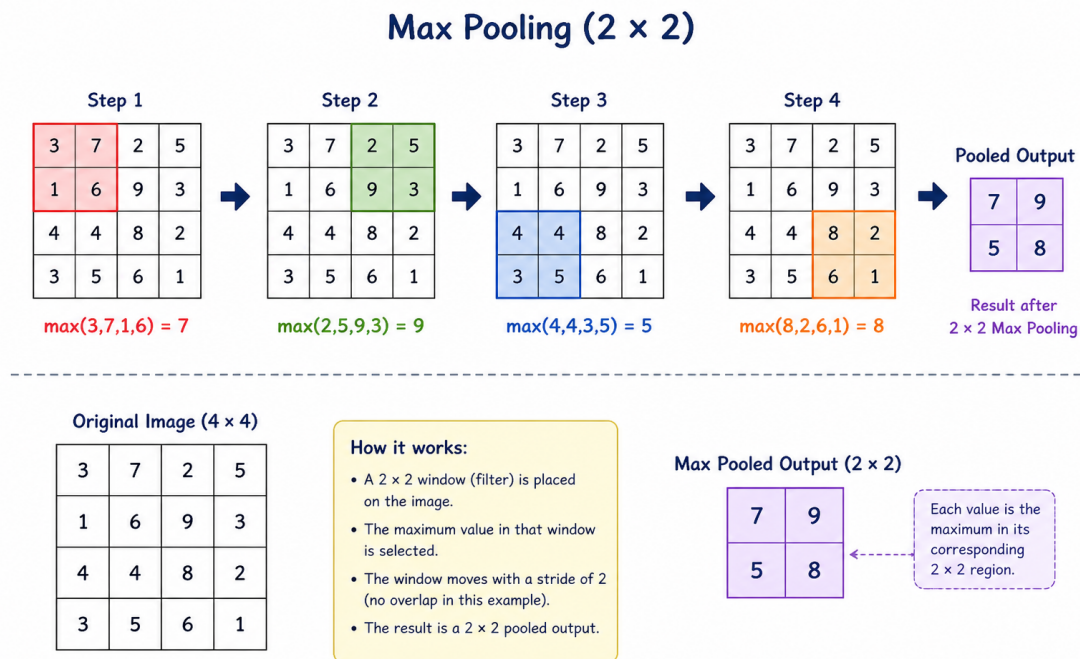


Figure 3.3: Example of 2×2 max pooling. The largest value from each local region is kept.

Pooling is useful for learning compact features, but it can remove small details. In segmentation, this is important because thin structures or small connections may disappear after downsampling. U-Net addresses this problem by combining downsampling for context with upsampling and skip connections for spatial recovery [1].

3.2 U-Net Segmentation Model

U-Net is an encoder–decoder CNN designed for pixel-level image segmentation. It extracts image context in the encoder, restores spatial resolution in the decoder, and uses skip connections to recover fine details [1]. This makes it useful for thin and connected structures such as vessels, roads, cracks, and biological objects [9].

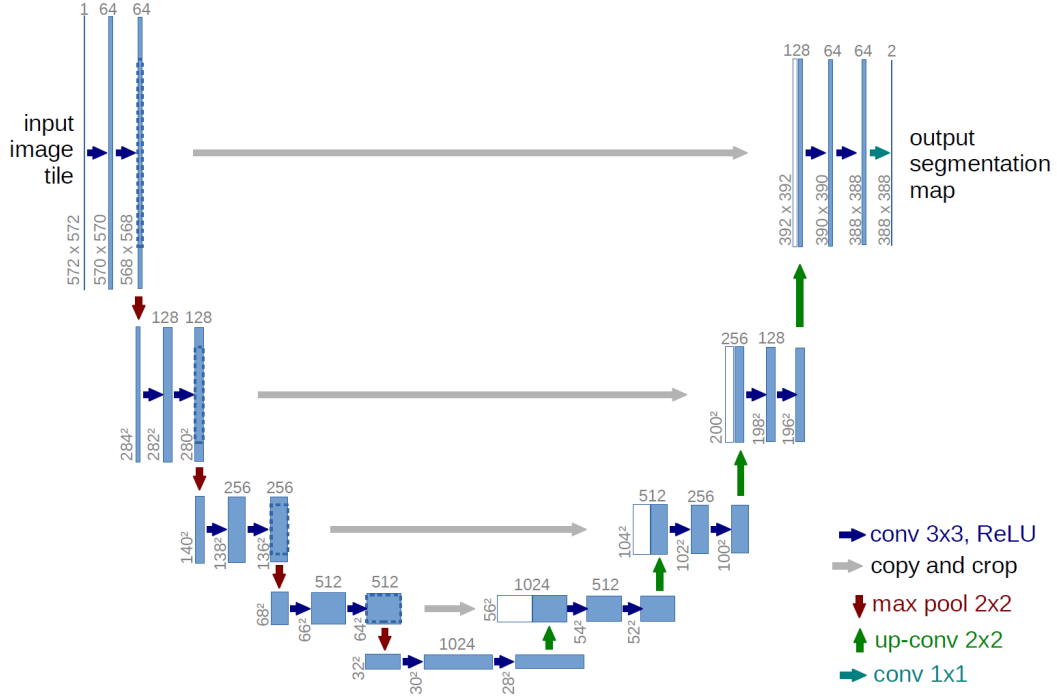


Figure 3.4: U-Net architecture with encoder, bottleneck, decoder, and skip connections [1].

3.2.1 Encoder. For an input image

$$x \in \mathbb{R}^{C \times H \times W},$$

the encoder applies convolution blocks to extract features:

$$C \times H \times W \rightarrow C' \times H \times W.$$

Max pooling then reduces the spatial size:

$$C' \times H \times W \rightarrow C' \times \frac{H}{2} \times \frac{W}{2}.$$

This helps the model learn larger image context.

3.2.2 Bottleneck and Decoder. The bottleneck is the deepest part of U-Net and contains the most compressed feature representation:

$$C_b \times \frac{H}{2^k} \times \frac{W}{2^k}.$$

The decoder then upsamples the features to recover spatial resolution:

$$H \rightarrow 2H, \quad W \rightarrow 2W.$$

This reconstructs the segmentation map at the original image size.

3.2.3 Skip Connections. Skip connections combine encoder and decoder features of the same spatial size. If the encoder feature is

$$C_s \times H \times W$$

and the decoder feature is

$$C_u \times H \times W,$$

their concatenation gives

$$(C_s + C_u) \times H \times W.$$

This helps recover fine boundaries and small structures that may be lost during downsampling [1].

3.2.4 Output Layer. A final 1×1 convolution maps decoder features to output classes. For binary segmentation,

$$P(x, y) \in [0, 1],$$

and thresholding gives the final mask. For multi-class segmentation with K classes, the model outputs

$$P_k(x, y), \quad k = 1, \dots, K,$$

and the predicted label is

$$\widehat{M}(x, y) = \arg \max_k P_k(x, y).$$

Although U-Net preserves spatial details, the final prediction is still guided by the loss function. Therefore, topology-aware losses are needed when connected components, branches, and holes must be preserved [2, 4, 6].

3.3 Loss Functions Used in the Framework

Loss functions guide the training of the segmentation model by measuring the difference between the prediction and the ground truth. In this work, the model predicts a probability map $P(x, y)$, while the ground-truth mask is denoted by $M(x, y)$. Since the target structures are often thin and connected, the loss should not only improve pixel accuracy, but also preserve overlap, connectivity, and topology [2, 3].

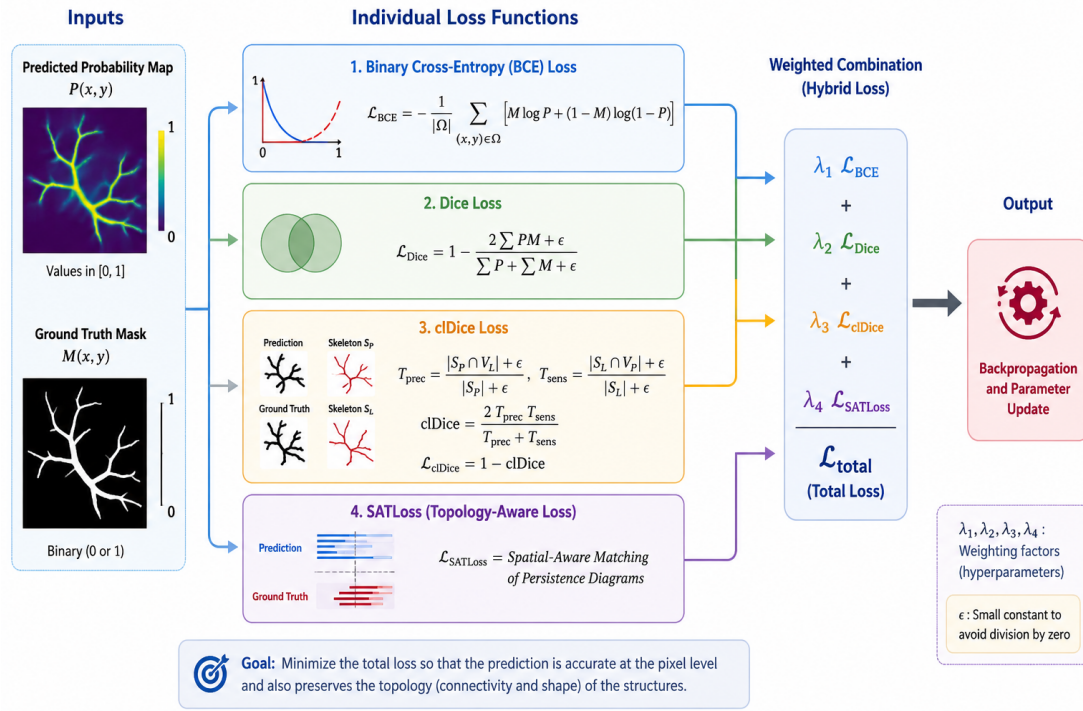


Figure 3.5: Hybrid loss framework. BCE improves pixel-wise classification, Dice improves region overlap, clDice supports connectivity, and SATLoss adds topology-aware supervision.

3.3.1 Binary Cross-Entropy Loss. Binary Cross-Entropy (BCE) compares each predicted pixel probability with its ground-truth label:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{|\Omega|} \sum_{(x,y) \in \Omega} [M(x, y) \log P(x, y) + (1 - M(x, y)) \log(1 - P(x, y))].$$

BCE is useful for pixel-level classification, but it treats pixels independently. Therefore, it does not directly penalize broken connections, missing branches, or false holes.

3.3.2 Dice Loss. Dice loss measures the overlap between the predicted segmentation and the ground truth:

$$\text{Dice}(P, M) = \frac{2 \sum P(x, y) M(x, y) + \epsilon}{\sum P(x, y) + \sum M(x, y) + \epsilon}, \quad \mathcal{L}_{\text{Dice}} = 1 - \text{Dice}(P, M).$$

Dice loss is important when the foreground is small compared with the background, as in roads, cracks, and vessels. However, good overlap does not always mean correct connectivity [2].

3.3.3 clDice Loss. clDice was introduced to improve connectivity preservation in thin and tubular structures [3]. It compares the skeleton of the prediction with the ground truth. Let S_P and S_M denote the skeletons of the prediction and ground truth, and let V_P and V_M denote their soft masks. Then

$$T_{\text{prec}} = \frac{|S_P \cap V_M| + \epsilon}{|S_P| + \epsilon}, \quad T_{\text{sens}} = \frac{|S_M \cap V_P| + \epsilon}{|S_M| + \epsilon}.$$

The cIDice score is

$$\text{cIDice} = 2 \frac{T_{\text{prec}} T_{\text{sens}}}{T_{\text{prec}} + T_{\text{sens}}}, \quad \mathcal{L}_{\text{cIDice}} = 1 - \text{cIDice}.$$

This loss helps preserve centerlines and connected paths, but it does not fully capture all topological features such as holes.

3.3.4 SATLoss. SATLoss uses persistent homology to compare topological features between prediction and ground truth [4, 6, 8]. Unlike ordinary persistence matching, it also includes spatial information, so features are matched not only by birth and death values but also by their image locations.

A spatial-aware matching cost can be written as

$$C_{\text{SA}}(p, q) = d_{\text{topo}}(p, q) + \lambda d_{\text{spatial}}(p, q),$$

where d_{topo} compares persistence values, d_{spatial} compares feature locations, and λ controls the effect of spatial distance. This reduces incorrect matching between features that look similar topologically but occur in different regions of the image.

3.3.5 Hybrid Loss Function. The main contribution of the training objective is to combine complementary losses so that the model learns pixel accuracy, region overlap, connectivity, and topology together. The hybrid loss is defined as

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{BCE}} + \lambda_2 \mathcal{L}_{\text{Dice}} + \lambda_3 \mathcal{L}_{\text{cIDice}} + \lambda_4 \mathcal{L}_{\text{SATLoss}}.$$

Each term has a clear role:

$$\begin{aligned} \mathcal{L}_{\text{BCE}} &\rightarrow \text{pixel classification}, & \mathcal{L}_{\text{Dice}} &\rightarrow \text{region overlap}, \\ \mathcal{L}_{\text{cIDice}} &\rightarrow \text{connectivity}, & \mathcal{L}_{\text{SATLoss}} &\rightarrow \text{topological consistency}. \end{aligned}$$

This combination is important because no single loss captures all requirements of topology-sensitive segmentation. BCE and Dice improve pixel and region accuracy, cIDice supports connected thin structures, and SATLoss adds global topological guidance. Therefore, the hybrid loss encourages predictions that are not only accurate in pixels but also structurally meaningful.

The weights λ_i can be manually set or regarded as learnable parameters in practice. The model can modify each loss term's contribution during training if they are learnable. This is helpful since too little topology supervision could not be able to repair structural errors, while too much might overconstrain the prediction.

3.4 Differentiability of Topological Loss

A loss function can train a neural network only if it provides gradients for backpropagation. BCE and Dice are differentiable from pixel and region values. Topological losses are different because

they depend on persistence diagrams, where each feature is controlled mainly by a **creator** cell and a **destroyer** cell [4, 6, 8].

Let θ be the network parameters. The hybrid loss is

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{BCE}} + \lambda_2 \mathcal{L}_{\text{Dice}} + \lambda_3 \mathcal{L}_{\text{clDice}} + \lambda_4 \mathcal{L}_{\text{SATLoss}},$$

so its gradient is

$$\frac{\partial \mathcal{L}_{\text{total}}}{\partial \theta} = \sum_{i=1}^4 \lambda_i \frac{\partial \mathcal{L}_i}{\partial \theta}.$$

Here, clDice is made differentiable through soft skeletonization [3], while SATLoss sends gradients mainly through creator and destroyer cells.

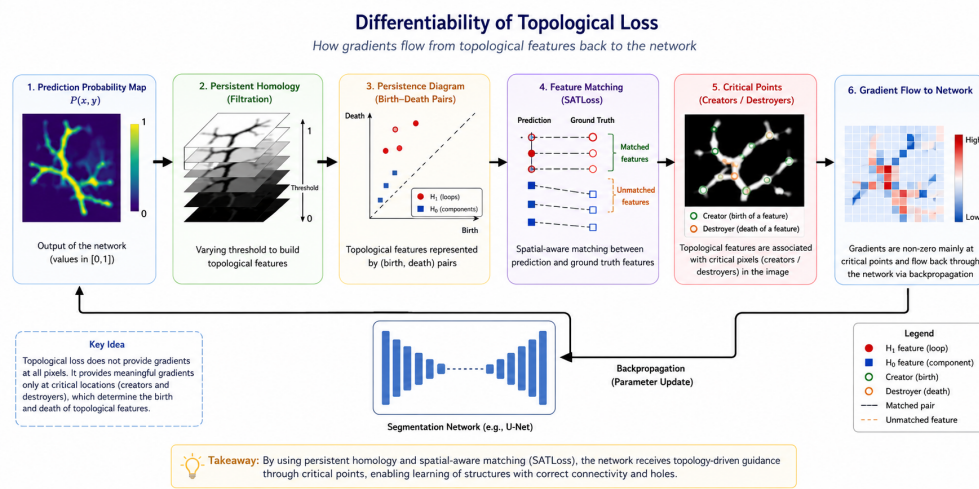


Figure 3.6: Gradients of topological loss mainly pass through creator and destroyer cells.

Lemma 3.1: (Critical-cell gradient).

Let L be the predicted probability map and let $p \in D(L)$ be a persistence point. If $c_b(p)$ is the creator cell and $c_d(p)$ is the destroyer cell, then

$$p = (L(c_b(p)), L(c_d(p))).$$

If p is matched to $\eta^*(p)$, the topological gradient can be written as

$$\nabla_{\theta} \mathcal{L}_{\text{topo}} = \sum_{p \in D(L)} 2s_p \left[(L(c_b(p)) - b(\eta^*(p))) \frac{\partial L(c_b(p))}{\partial \theta} + (L(c_d(p)) - d(\eta^*(p))) \frac{\partial L(c_d(p))}{\partial \theta} \right].$$

Here, $\eta^*(p)$ is the matched target point or diagonal projection, and s_p is determined by the matching process [4, 6, 8]. This formula shows that the loss compares predicted birth and death values with the matched target values, then updates mainly the creator and destroyer cells.

The gradient in Lemma 3.1 is valid only when the underlying topological structure remains unchanged. Therefore, it is important to understand when the topological loss is differentiable and when differentiability may fail.

Differentiability conditions.

The topological loss is locally differentiable when the filtration order, creator–destroyer cells, persistence pairing, and matching remain fixed under a small change in the prediction map. In this stable case, the loss behaves like an ordinary differentiable function of the critical pixel values.

However, the loss may fail to be differentiable when a small change changes a discrete topological choice. This can happen when

$$p_i = p_j,$$

when a creator or destroyer changes,

$$b_k = p_5 \longrightarrow b_k = p_7,$$

when the persistence pairing changes, or when a point switches between a target point and the diagonal. Therefore, topological loss is piecewise differentiable: differentiable in stable regions, but possibly non-differentiable at switching boundaries.

Example 3.3: One gradient update.

Suppose

$$L(c_b(p)) = 0.80, \quad b(\eta^*(p)) = 0.90.$$

The creator error is -0.10 , so gradient descent increases the creator value. Also suppose

$$L(c_d(p)) = 0.20, \quad d(\eta^*(p)) = 0.10.$$

The destroyer error is 0.10 , so gradient descent decreases the destroyer value. With learning rate $\eta = 0.1$,

$$L(c_b(p))_{\text{new}} = 0.80 - 0.1(-0.10) = 0.81,$$

and

$$L(c_d(p))_{\text{new}} = 0.20 - 0.1(0.10) = 0.19.$$

Thus, the creator moves toward the target birth value and the destroyer moves toward the target death value. SATLoss extends this by using spatial-aware matching, so important structures are encouraged to appear in the correct image regions [8].

4. Implementation and Results

This chapter presents the implementation and experimental results of the proposed topology-preserving segmentation framework. The experiments were carried out on four topology-sensitive datasets: road networks, cracks, DRIVE retinal vessel, and *C. elegans* microscopy images. These datasets contain thin and connected structures, making them suitable for evaluating both pixel-wise accuracy and topological consistency. The chapter describes the preprocessing steps, training setup, evaluation metrics, and experimental results for each dataset.

4.1 Dataset Description

4.1.1 Road Dataset. The road dataset was obtained from the Massachusetts Roads Dataset, introduced by Mnih for aerial image labeling and road extraction [14]. The original images are RGB satellite images of size approximately 1500×1500 , with binary masks indicating road and non-road regions. This dataset is important because road networks contain branches, intersections, and long connected structures.

4.1.2 *C. elegans* Dataset. The *C. elegans* dataset was obtained from the Broad Bioimage Benchmark Collection, which provides annotated microscopy image sets for biological image analysis [15]. In this work, the original grayscale microscopy images have size 696×520 , with corresponding binary foreground masks. This dataset was used for transfer learning from the road segmentation model to biomedical image segmentation.

4.1.3 Crack Dataset. The crack dataset was obtained from the CrackTree dataset, introduced for automatic crack detection from pavement images [16]. The original crack images have size 256×256 , with corresponding binary crack masks. A total of 7247 valid image-mask pairs were used. Crack segmentation is topology-sensitive because cracks are thin, irregular, and easily broken during prediction.

4.1.4 DRIVE Retinal Vessel Dataset. The DRIVE retinal vessel dataset was obtained from the Digital Retinal Images for Vessel Extraction (DRIVE) database [17]. The dataset consists of retinal fundus images together with manually annotated vessel masks for retinal vessel segmentation. The original RGB images have size 584×565 . Retinal vessel segmentation is topology-sensitive because blood vessels form thin connected networks that are easily broken by segmentation errors. In this work, the dataset was used as an additional transfer learning experiment to evaluate whether topology-aware features learned from road segmentation could generalize to retinal vessel structures.

4.2 Data Preprocessing

Before training, all datasets were prepared using a common preprocessing pipeline. Each image was first paired with its corresponding ground-truth mask, and the split into training, validation,

and testing sets was performed before patch extraction. This was done to avoid data leakage, since patches from the same original image should not appear in different subsets.

All masks were converted into binary form:

$$0 = \text{background}, \quad 1 = \text{foreground}.$$

The images and masks were then divided into patches of size 96×96 . For the road, C. elegans, and DRIVE datasets, overlapping patches were extracted using a stride of 48. Image values were normalized from $[0, 255]$ to $[0, 1]$, and the tensors were rearranged into channel-first format for PyTorch training:

$$H \times W \times C \rightarrow C \times H \times W.$$

Although the general preprocessing steps were similar, some dataset-specific operations were required. For the road dataset, patch filtering was applied because road pixels occupy only a small portion of the satellite images. After filtering, the final road dataset contained:

381913 training patches, 79316 validation patches, 81907 testing patches.

For the C. elegans dataset, the original grayscale microscopy images of size 696×520 were converted into three-channel inputs to match the pretrained road model. After patch extraction, 6103 image-mask patch pairs were obtained.

For the crack dataset, the original RGB images of size 256×256 were cropped into 96×96 patches. Border patches were padded when necessary, and mild mask cleaning was applied by removing small isolated objects and holes. The final crack dataset contained:

26356 training patches, 5718 validation patches, 5620 testing patches.

For the DRIVE retinal vessel dataset, field-of-view masks were used to remove patches outside the retinal region. Patches with very small vessel content were also filtered out. The final DRIVE dataset contained:

1301 training patches, 278 validation patches, 280 testing patches.

4.3 Training Procedure

The experiments were implemented in PyTorch using the U-Net architecture. All datasets were trained using image patches, and the model produced probability maps that were compared with the corresponding binary masks. The general implementation flow is summarized in Algorithm 4.1.

Table 4.1: Pseudocode for topology-aware segmentation training

Algorithm 4.1: Topology-aware segmentation training pipeline**Input:** Images $\mathcal{I}$, masks $\mathcal{M}$, loss configuration $\mathcal{L}$, threshold $\tau = 0.5$ **Output:** Best trained model and evaluation metrics

-
1. Pair each image $I_i \in \mathcal{I}$ with its corresponding mask $M_i \in \mathcal{M}$.
 2. Split the paired data into training, validation, and testing sets.
 3. Extract image-mask patches and normalize image values to $[0, 1]$.
 4. Initialize the U-Net model with parameters θ .
 5. **for** each epoch $e = 1, \dots, E$ **do**
 6. **for** each mini-batch (I_b, M_b) in the training set **do**
 7. Compute the probability map $P_b = f_\theta(I_b)$.
 8. Compute the selected loss $\mathcal{L}(P_b, M_b)$.
 9. Update θ using the Adam optimizer.
 10. **end for**
 11. Evaluate the model on the validation set.
 12. Save the model if the validation performance improves.
 13. **end for**
 14. Predict probability maps on the test set using the best model.
 15. Convert probabilities to binary masks: $\widehat{M} = 1$ if $P \geq \tau$, otherwise $\widehat{M} = 0$.
 16. Compute Accuracy, Dice, cIDice, β_0 error, and β_1 error.
-

4.3.1 Road and Crack Segmentation Training. The road and crack datasets were trained using the same framework because both contain thin and connected structures. Six loss configurations were tested to compare ordinary pixel-wise learning with topology-aware learning.

- | | |
|--|--|
| 1. BCE Only: | $\mathcal{L}_{\text{BCE}}$ |
| 2. BCE + SATLoss Creator: | $\mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{SATLoss}}^{\text{creator}}$ |
| 3. BCE + SATLoss Creator-Destroyer: | $\mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{SATLoss}}^{\text{creator+destroyer}}$ |
| 4. Hybrid Loss: | $\mathcal{L}_{\text{Hybrid}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{clDice}}$ |
| 5. Hybrid + SATLoss Creator: | $\mathcal{L}_{\text{Hybrid}} + \mathcal{L}_{\text{SATLoss}}^{\text{creator}}$ |
| 6. Hybrid + SATLoss Creator-Destroyer: | $\mathcal{L}_{\text{Hybrid}} + \mathcal{L}_{\text{SATLoss}}^{\text{creator+destroyer}}$ |

The BCE model was used as the baseline. The Hybrid model combined pixel accuracy, region overlap, and connectivity preservation. The SATLoss models added topology-aware supervision through creator cells or through both creator and destroyer cells. To allow the model to modify the level of topological supervision during training, the SATLoss weight was used as a learnable parameter in the road experiments.

4.3.2 Transfer Learning on C. elegans and DRIVE. The DRIVE and C. elegans datasets were subjected to transfer learning. Rather than starting from scratch, the model was initialized using pretrained road segmentation weights. The Hybrid + SATLoss Creator setup worked in the transferred model:

$$\mathcal{L}_{\text{Hybrid}} + \mathcal{L}_{\text{SATLoss}}^{\text{creator}}.$$

This model was selected because it provided a stable balance between structural preservation and visual prediction quality.

Fine-tuning was performed in two stages. First, the model was trained using

$$\mathcal{L}_{\text{Hybrid}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{clDice}}.$$

Then SATLoss was added to restore topology-aware supervision:

$$\mathcal{L}_{\text{Hybrid}} + \mathcal{L}_{\text{SATLoss}}.$$

This made it possible for the model to transfer structural information learned from road networks to biomedical structures such as retinal vessels and worm bodies.

4.4 Evaluation Metrics

Both pixel-level and topology-aware criteria were used to evaluate the models. Segmentation quality was measured using accuracy, dice, and clDice, while topological agreement was measured using β_0 and β_1 errors.

The percentage of pixels that are correctly categorized is known as accuracy. The overlap between the ground truth and the prediction is measured by dice. Skeleton-level connectivity and continuity were evaluated using clDice since the datasets contain thin connected structures [3].

Topology was evaluated using Betti number errors. The β_0 error measures connected-component mistakes, while the β_1 error measures hole-related mistakes:

$$\beta_0 \text{ error} = |\beta_0(P) - \beta_0(M)|, \quad \beta_1 \text{ error} = |\beta_1(P) - \beta_1(M)|.$$

Better segmentation performance is shown by higher Accuracy, Dice, and clDice. Better preservation of connected components and holes is shown by lower β_0 and β_1 errors. These measures were combined since proper topology is not always ensured by good pixel accuracy alone.

4.5 Experimental Results

4.5.1 Road Dataset Results. Pixel-wise training and topology-aware training were compared using the road dataset. BCE, Dice, clDice, and SATLoss were used to evaluate six models. The objective was to determine whether topology-aware supervision enhances structural consistency and road connection.

Table 4.2: Quantitative results for road segmentation. Higher Accuracy, Dice, and cIDice are better, while lower β_0 and β_1 errors are better.

Model	Loss	Acc	Dice	cIDice	β_0	β_1
BCE Only	0.1331	0.9545	0.7204	0.7310	0.7322	0.1150
BCE + SAT-C	0.1360	0.9543	0.7182	0.7291	0.7534	0.1062
BCE + SAT-CD	0.1318	0.9542	0.7178	0.7292	0.7440	0.1014
Hybrid	0.1681	0.9524	0.7207	0.7471	0.4688	0.1146
Hybrid + SAT-C	0.1601	0.9517	0.7209	0.7455	0.4916	0.0998
Hybrid + SAT-CD	0.1601	0.9517	0.7209	0.7455	0.4916	0.0998

The BCE baseline had the highest accuracy, 0.9545, although it had a large β_0 error, 0.7322. This shows that connected road structures are not ensured by high pixel precision alone. The Hybrid model achieved the best cIDice score of 0.7471 and reduced the β_0 error from 0.7322 to 0.4688. This indicates better preservation of road connectivity.

Explicit topology-aware supervision was introduced by adding SATLoss. The Hybrid + SATLoss Creator and Hybrid + SATLoss Creator–Destroyer models had the lowest β_1 error, 0.0998 and the best Dice score, 0.7209. Additionally, their cIDice score of 0.7455 was very similar to the Hybrid model. This shows how SATLoss improved hole-related topology while preserving competitive segmentation quality.

Rather than being manually set, the SATLoss weight was treated as a learnable parameter in these last road tests. As a result, throughout training, the model was able to modify the level of topological supervision. Then, using the same Dice, cIDice, β_0 , and β_1 parameters as the Creator-only model, the Creator–Destroyer model achieved stable predictions. This implies that learnable topology weighting contributed in finding a balance between the quality of visual segmentation and structural preservation.

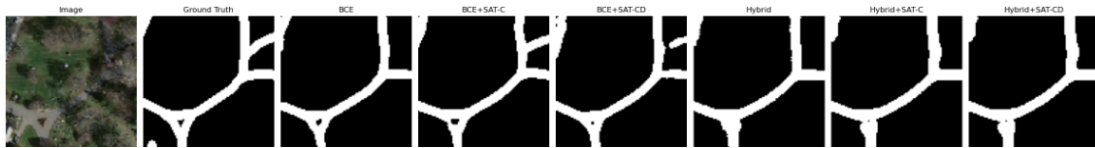


Figure 4.1: The six road segmentation models are compared qualitatively. Compared to BCE-only training, the topology-aware models more reliably preserve connections and road continuity.

Figure 4.1 shows that BCE captures many road pixels, but topology-aware models preserve road continuity and junctions more consistently. Figure 4.2 shows a representative Hybrid + SAT-Loss Creator–Destroyer prediction, where the main road structure follows the ground truth while remaining visually stable.

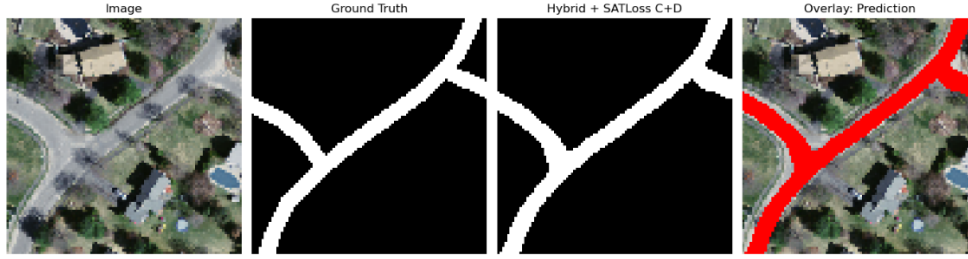


Figure 4.2: Representative prediction from the Hybrid + SATLoss Creator–Destroyer model. The model preserves major road structures and intersections while maintaining stable topology-aware segmentation.

Overall, the road results show the benefit of topology-aware learning over BCE-only training. BCE produced strong pixel accuracy, but it had larger connected-component errors. In contrast, the Hybrid and SATLoss-based models improved structural metrics, especially cIDice, β_0 error, and β_1 error. Although the differences between the Hybrid and SATLoss-based models were small, SATLoss added explicit topological guidance and the learnable weight made this guidance more stable during prediction.

4.5.2 Crack Dataset Results. The crack dataset was used to evaluate the effect of topology-aware learning on thin and irregular crack structures. Similar to the road experiments, six models were trained using different combinations of BCE, Dice, cIDice, and SATLoss. The aim was to determine whether topology-aware supervision improves crack continuity and structural preservation.

Table 4.3: Quantitative results for crack segmentation. Higher Accuracy, Dice, and cIDice are better, while lower β_0 and β_1 errors are better.

Model	Acc	Dice	cIDice	β_0	β_1	Threshold
BCE Only	0.9042	0.6308	0.6002	1.7445	0.1461	0.3617
BCE + SAT-C	0.9061	0.6283	0.5949	1.5852	0.1564	0.3535
BCE + SAT-CD	0.9055	0.6268	0.5973	1.3941	0.1313	0.3570
Hybrid	0.9295	0.7127	0.7410	0.3251	0.0498	0.5781
Hybrid + SAT-C	0.9057	0.6332	0.6468	1.2641	0.0808	0.4922
Hybrid + SAT-CD	0.9197	0.6752	0.7012	0.5664	0.0573	0.5551

The BCE-only baseline achieved Accuracy 0.9042, Dice 0.6308, and cIDice 0.6002. However, it also produced a very large β_0 error of 1.7445, indicating many disconnected crack components in the predicted segmentation. This shows that pixel-wise learning alone is not sufficient for preserving thin crack connectivity.

The Hybrid model, which combines BCE, Dice, and cIDice losses, achieved the best overall performance. It obtained the highest Accuracy (0.9295), Dice (0.7127), and cIDice (0.7410), together with the lowest β_0 error (0.3251) and β_1 error (0.0498). Compared with BCE-only training, the Hybrid model reduced the β_0 error from 1.7445 to 0.3251, showing a major improvement in connected crack preservation and structural continuity.

The SATLoss-based models also improved topology compared with BCE-only training. In particular, the Hybrid + SATLoss Creator–Destroyer model achieved Dice 0.6752, cIDice 0.7012, β_0 error 0.5664, and β_1 error 0.0573. Although these values were lower than the Hybrid-only model, they still represent a strong improvement over BCE-only segmentation. The reduction in β_0 error from 1.7445 to 0.5664 indicates that topology-aware supervision helped reduce disconnected crack regions.

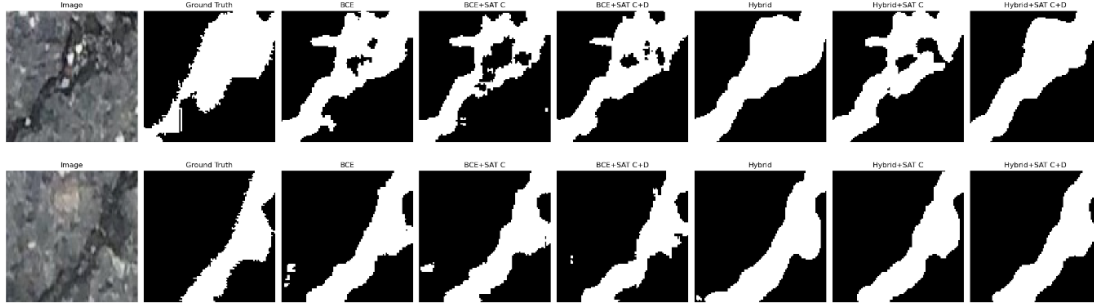


Figure 4.3: Qualitative comparison of crack segmentation results across all trained models. The topology-aware models preserve crack continuity more consistently than BCE-only training.

The qualitative results in Figure 4.3 support the numerical observations. BCE-only predictions contain fragmented and noisy crack structures, especially around thin boundaries. In contrast, the Hybrid and SATLoss-based models generally produce smoother and more continuous crack predictions.

However, unlike the road dataset, adding SATLoss to the Hybrid model did not produce the best final quantitative performance. This may be because crack images contain highly irregular and noisy structures, making strong topology constraints more sensitive to uncertain regions during training. Consequently, the Hybrid model provided the best balance between overlap accuracy, connectivity preservation, and segmentation stability for this dataset.

Overall, the crack experiments demonstrate the importance of structural learning in thin-object segmentation. The Hybrid and topology-aware models significantly reduced topological errors compared with BCE-only training, especially in terms of cIDice and β_0 error. These results show that preserving connectivity is essential for crack segmentation, since small disconnected regions can strongly affect the final structural interpretation of the cracks.

4.5.3 C. elegans Transfer Learning Results. The *C. elegans* experiment was used to evaluate whether topology-aware features learned from road segmentation could transfer to biomedical microscopy images. The road model was first tested before fine-tuning, and then adapted to the *C. elegans* dataset using transfer learning.

Table 4.4: Quantitative results for *C. elegans* transfer learning.

Model	Acc	Dice	cIDice	β_0	β_1
Road model before fine-tuning	0.8960	0.1604	–	–	–
Transfer learning	0.9539	0.7449	0.7727	0.3904	0.0950
Transfer learning + SATLoss Creator	0.9539	0.7449	0.7727	0.3904	0.0950

Before fine-tuning, the pretrained road model did not adapt well to the microscopy images, giving a low Dice score of 0.1604. After transfer learning, the model improved strongly, reaching Accuracy 0.9539, Dice 0.7449, and cDice 0.7727. The β_0 error of 0.3904 and β_1 error of 0.0950 also show that the model preserved the main connected worm structures with few topological mistakes.

The transfer learning and Transfer learning + SATLoss Creator results were the same numerically in this experiment. This suggests that the pretrained topology-aware representation had already transferred useful structural information during fine-tuning. The SATLoss Creator stage still remains important because it keeps the training objective aligned with topology preservation.

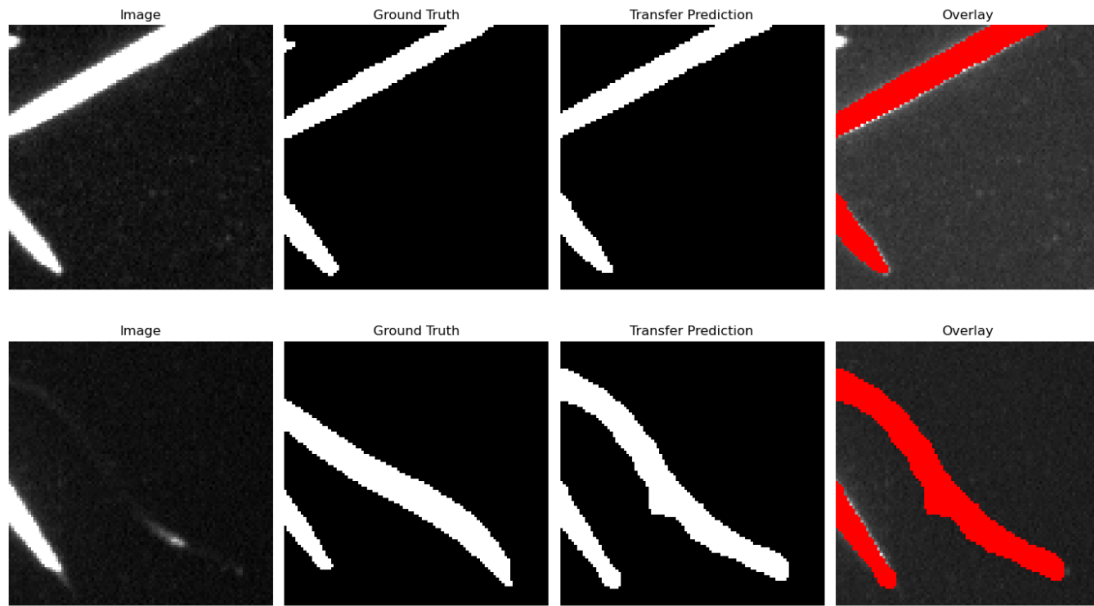


Figure 4.4: Representative *C. elegans* transfer learning predictions. The model preserves thin and connected worm structures while maintaining good agreement with the ground-truth masks.

The qualitative results in Figure 4.4 show that the predicted masks follow the main worm structures and remain close to the ground truth. This supports the numerical results, especially the high cDice score, which indicates good preservation of connected structures.

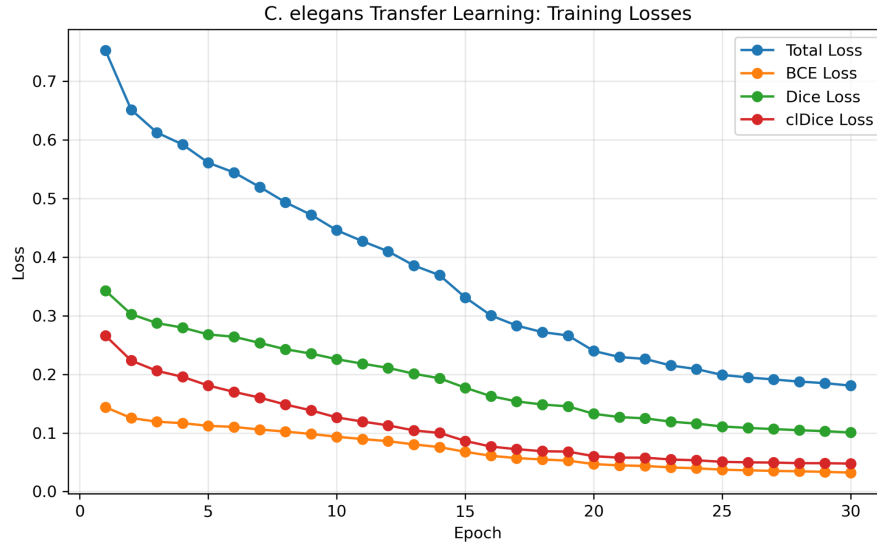


Figure 4.5: Training losses during *C. elegans* transfer learning. The decreasing BCE, Dice, and cIDice losses indicate stable fine-tuning.

Overall, the *C. elegans* experiment shows that topology-aware features learned from road segmentation can generalize to biomedical microscopy images. After fine-tuning, the model preserved thin connected worm patterns much better than the original road model before adaptation.

4.5.4 DRIVE Retinal Vessel Results. The DRIVE experiment was used to evaluate whether topology-aware transfer learning could generalize from road segmentation to retinal vessel extraction. This dataset is suitable because retinal vessels are thin, connected, and highly sensitive to broken branches.

Table 4.5: Quantitative results for DRIVE retinal vessel transfer learning.

Model	Acc	Dice	cIDice	β_0	β_1	Threshold
Road model before fine-tuning	0.8645	0.0000	–	–	–	0.5000
DRIVE Transfer Learning	0.9518	0.7976	0.7828	1.4179	0.7107	0.5000
DRIVE Transfer Learning + SATLoss Creator	0.9518	0.7976	0.7828	1.4179	0.7107	0.5000

Before fine-tuning, the pretrained road model failed to segment the retinal vessels correctly, giving Dice 0.0000. After transfer learning, the model improved strongly, reaching Accuracy 0.9518, Dice 0.7976, and cIDice 0.7828. This shows that the model adapted from road structures to retinal vessel structures after fine-tuning.

The transfer learning and Transfer Learning + SATLoss Creator models produced the same numerical results in this experiment. This suggests that the transferred topology-aware representation already captured useful connectivity information during fine-tuning. The high cIDice score shows that the model preserved vessel continuity rather than only improving pixel-wise appearance.

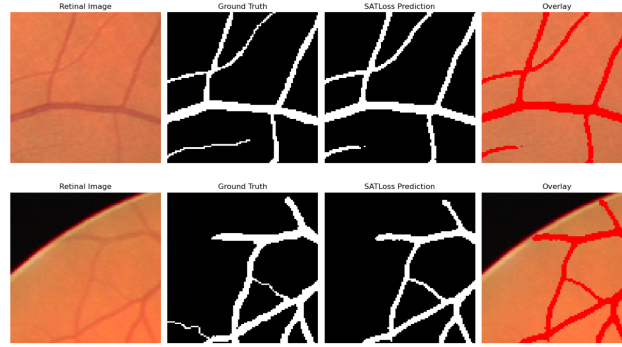


Figure 4.6: Representative retinal vessel patch predictions. The predicted vessels follow the ground-truth connectivity and align well with the retinal image structures.

To further evaluate large-scale vessel connectivity, full-image inference was also performed. Since the model was trained on patches, the full retinal image was divided into patches, each patch was predicted separately, and the patch probabilities were merged to obtain the final full-image prediction.

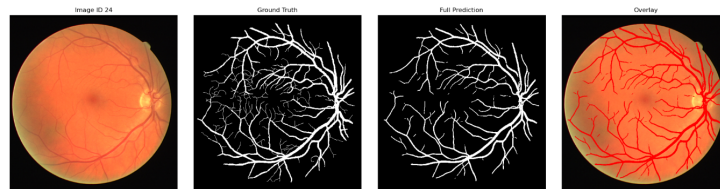


Figure 4.7: Full retinal vessel prediction using topology-aware transfer learning. The model preserves the main vessel tree and captures many thin vessel branches.

For the representative full-image result, the model achieved Accuracy 0.9720, Dice 0.8196, and cLDice 0.8511. The high cLDice value indicates strong preservation of vessel connectivity and branching structure. Although the β_0 and β_1 errors were larger on the full image, this is expected because retinal vessel networks contain many small branches and loops across the whole image.

Overall, the DRIVE experiment shows that topology-aware features learned from road segmentation can transfer to retinal vessel segmentation. After fine-tuning, the model preserved connected vessel structures and branching patterns, which are important for retinal image analysis.

5. Conclusion and Future Work

5.1 Conclusion

This thesis studied topology-preserving image segmentation using spatial-aware persistent feature matching. The aim was to produce segmentation results that are accurate at the pixel level and also preserve important structures such as connected components, branches, holes, and continuity.

The main contribution of this work is the development of a topology-aware segmentation framework that combines U-Net with BCE, Dice, cLDice, and SATLoss. BCE and Dice improve pixel classification and region overlap, cLDice supports connectivity preservation, and SATLoss introduces persistent-homology-based structural guidance. Spatial-aware matching further improves this idea by considering both the persistence values and the image locations of topological features.

The road dataset results showed why topology-aware learning is important. BCE-only training achieved high pixel accuracy, but it produced larger connected-component errors. In contrast, the Hybrid model reduced β_0 error from 0.7322 to 0.4688, while the Hybrid + SATLoss models achieved the lowest β_1 error of 0.0998. This shows that adding connectivity-aware and topology-aware losses improved structural preservation beyond pixel accuracy.

This observation was also validated by the crack dataset. With a significant β_0 error of 1.7445, BCE-only training showed several disconnected crack components. The best Dice and cLDice scores were obtained by the Hybrid model, which also decreased this inaccuracy to 0.3251. This shows that the best balance for noisy crack architectures was obtained by combining BCE, Dice, and cLDice. When compared to BCE-only training, the SATLoss-based models continued to reduce topological errors; however, the results showed that strong topology constraints need to be carefully balanced for irregular and noisy pictures.

Another contribution of this work is the use of learnable SATLoss weighting. In the road experiments, the SATLoss weight was treated as a learnable parameter, allowing the model to adjust the strength of topology supervision during training. This helped the Creator–Destroyer model produce stable predictions while maintaining topology-aware guidance.

The transfer learning experiments on *C. elegans* and DRIVE retinal vessels showed that topology-aware features learned from road segmentation can generalize to biomedical images. After fine-tuning, the pretrained road model improved from Dice 0.1604 to 0.7449 on *C. elegans*, and from Dice 0.0000 to 0.7976 on DRIVE. The high cLDice scores in both datasets showed that the model preserved thin connected structures rather than only learning pixel appearance.

In general, the findings show that topology-aware learning provides helpful structural guidance for problems requiring thin and connected objects. The suggested framework supports predictions that are closer to the ground truth topology and have structural importance in addition to aiming for greater pixel accuracy.

5.2 Limitations

There are a number of limitations to this study. First, because persistent homology and SATLoss require additional topological computations, topology-aware training is computationally more costly than traditional pixel-wise training. Second, not all measures are improved by topology constraints. For instance, in the crack dataset, the Hybrid model did better than the SATLoss models, indicating that strong topological supervision might be sensitive to noisy and irregular structures. Third, the results might not fully apply to multi-class segmentation, three-dimensional images, or other deep learning architectures without more testing because the studies mostly used U-Net and binary segmentation. Lastly, even though learnable SATLoss weighting increased stability, it is still difficult to achieve the right balance between topology preservation and pixel accuracy.

5.3 Future Work

Future work can extend this study to larger datasets, multi-class segmentation, and three-dimensional medical images. More efficient implementations of persistent homology and SATLoss can also be explored to reduce training time and memory cost.

Other segmentation architectures, such as Attention U-Net, UNet++, and Transformer-based models, can also be tested with topology-aware losses. In addition, adaptive weighting strategies can be further studied to automatically balance BCE, Dice, cIDice, and SATLoss during training.

Finally, this framework can be applied to other topology-sensitive problems such as neuron tracing, airway segmentation, river extraction, and biomedical vessel analysis.

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